

Adroit Technologies

Protocol Driver

Name	Adroit Generic Email SMTP
DLL	EMAIL



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Contents

1. Overview	4
2. Wiring	5
3. Device Configuration	6
3.1. Email Configuration	6
3.1.1. Outgoing Mail Settings	6
3.2. MACRO Usage in Email	6
4. Supported Addresses	8
4.1. Outgoing Email Rawbytes Slots.	8
5. Protocol Definition and Considerations	9
6. General Notes and Troubleshooting	10
6.1. Contact Details.....	10
6.2. Adroit Technologies Knowledge Base.....	10
6.3. Troubleshooting	10
7. Appendix	13
7.1. Appendix A: Gmail Setup for Email Drivers using IMAP [New Google Security Policy].....	13
8. Revisions.....	18
8.1. Document Revision.....	18
8.2. DLL Revision	18

1. Overview

This driver is used to send email messages via Simple Mail Transfer Protocol (SMTP), designed specifically for sending alarms or any other information such as Adroit tag values, SV and PV values, as well as attachments.

The driver currently only supports RAWBYTES slots, of which only the Notify Agent currently provides a suitable Rawbytes slot.

The Notify Agent allows for manual email sending, and it also exposes the various sub items such as TO, CC, FROM, SUBJECT and so forth as individual VARSTRING slots so an email interface could easily be crafted onto a mimic.

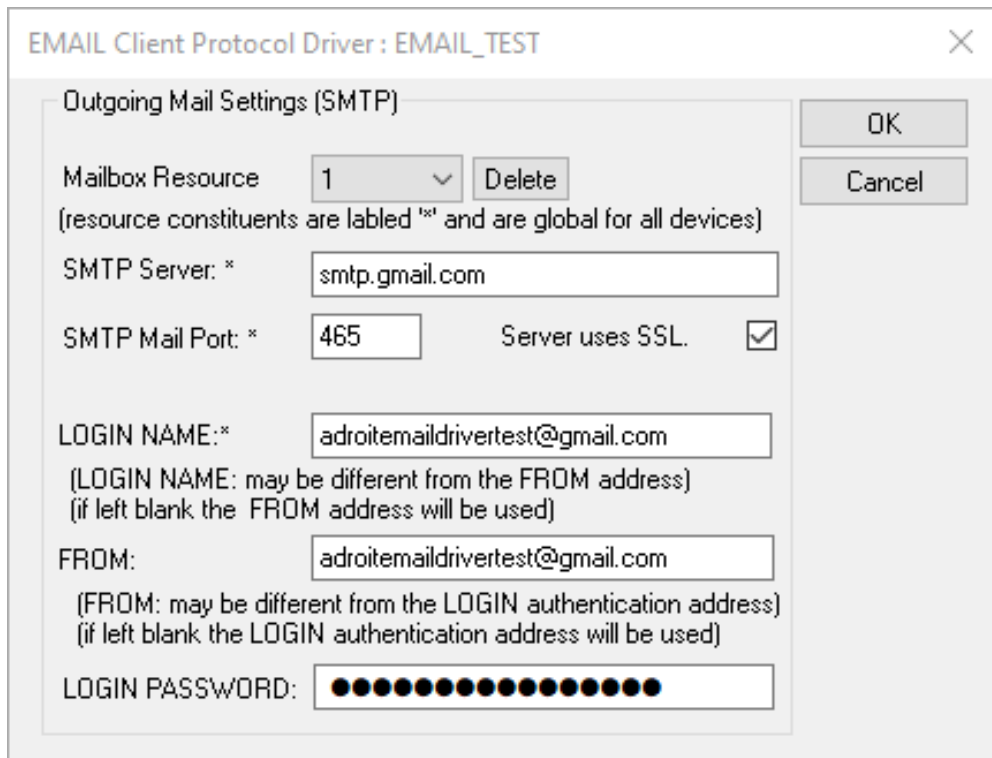
The subject line and body may contain text macros that are parsed to allow for the additional information from Adroit, specifically slot and alarm details, to be sent if required and not just a simple alarm text field.

2. Wiring

Not applicable.

3. Device Configuration

3.1. Email Configuration



EMAIL Client Protocol Driver : EMAIL_TEST

Outgoing Mail Settings (SMTP)

Mailbox Resource: 1 [Delete]
(resource constituents are labeled '*' and are global for all devices)

SMTP Server: * smtp.gmail.com

SMTP Mail Port: * 465 Server uses SSL. ☒

LOGIN NAME: * adroitemaildrivertest@gmail.com
(LOGIN NAME: may be different from the FROM address)
(if left blank the FROM address will be used)

FROM: adroitemaildrivertest@gmail.com
(FROM: may be different from the LOGIN authentication address)
(if left blank the LOGIN authentication address will be used)

LOGIN PASSWORD: [Masked]

OK Cancel

3.1.1. Outgoing Mail Settings

Mailbox Resource – A specific resource ID that uniquely identifies an email resource. The relevant mailbox settings are saved and remembered for the next time the configuration is brought up. Additional Adroit devices can be created and quickly configured with existing mailbox resources. A prompt will appear when settings have been changed that do not impact the resource constituents whether it must affect all existing devices that use that resource or if it must create a new resource to not affect current devices. For this email driver, this only applies to the LOGIN PASSWORD field.

From – The fully qualified email address the email will be sent from.

Login Password – The account domain password.

SMTP Server – The specific SMTP server to contact according to protocol specifications. Usually matches the machine name e.g. MAIL.adroit.co.za

SMTP Mail Port – Port required on the mail server for SMTP. Typically, this is 465 for SSL. Other ports could include 25 and 587.

3.2. MACRO Usage in Email

This email driver is used in conjunction with the Notify agent.

<ALM> will insert the alarm string received from the Notify Agent.



PLEASE NOTE

The text that will be substituted into the <ALM> macro first needs to be generated by an actual alarm. If you see this field as blank in testing, it means an alarm has not been triggered to generate it.

<Agent.slot> will insert the result of a string query to the agent slot. For example, an Analog.value slot will result in the engineering value being inserted.

Emails manually sent will contain the macros without the substitution for verification. The macros will only be expanded through the traditional Adroit alarming mechanism. Macros can be placed anywhere in the text fields of SUBJECT or BODY where duplicates are allowed.

The ATTACHMENT line may only send 1 file at a time, though any file may be sent. For example, this can be inserted into the BODY:

Hello Fred

An alarm for your attention: (<ALM>),

The dam level is at <DAM_LVL.VALUE> meters.

The pump is <RUN_STATE.TEXT>.

Goodbye from Adroit

4. Supported Addresses

4.1. Outgoing Email Rawbytes Slots.

The driver address is "EMAIL.x" where $\{x \geq 0, X \leq 250\}$. When scanning, note the following:

- Must be **Output enabled**.
- Must be **Input disabled**.
- Must be scanned to a Notify agent's **Email-rawValue** slot.

The other Notify agent slots are internal and should not be directly scanned to any driver.

Noting that the FROM field is defined once in the device setup, a system may require more. This can be achieved by adding as many mode "devices" as needed.

If the FROM field is not significant, up to 250 Notify agent slots may be scanned to one device by incrementing the scan address index.

5. Protocol Definition and Considerations

```
ASCII DATASCOPE for EMAIL_TEST on Port: 465

[Result= Connected Successfully!]

2022-12-09 16:10:05 - To: adrdrivertest@outlook.com
Subject: Email Notification Test 1
Body: This is an example email.
[ANALOG1.value = 565.00000]

Dir0, RSET

Dir1, 250 2.1.5 Flushed pk26-20020a170906d7ba00b007c0dcb30103sm597911ejb.103 - gsmt
Dir0, MAIL FROM: <adroitemaildrivertest@gmail.com>

Dir1, 250 2.1.0 OK pk26-20020a170906d7ba00b007c0dcb30103sm597911ejb.103 - gsmt
Dir0, RCPT TO:<adrdrivertest@outlook.com>

Dir1, 250 2.1.5 OK pk26-20020a170906d7ba00b007c0dcb30103sm597911ejb.103 - gsmt
Dir0, DATA

Dir1, 354 Go ahead pk26-20020a170906d7ba00b007c0dcb30103sm597911ejb.103 - gsmt
Dir0, .

Dir1, 250 2.0.0 OK 1670595007 pk26-20020a170906d7ba00b007c0dcb30103sm597911ejb.103 - gsmt

[Result= Message Sent Successfully!]

Dir0, QUIT

Dir1, 221 2.0.0 closing connection pk26-20020a170906d7ba00b007c0dcb30103sm597911ejb.103 - gsmt
Dir2, Disconnected.
```

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

1. When I try send a test email in the Notify agent with an <ALM> macro, I get the following printed in my email body: **Alarm trigger required to generate ALM message. Placeholding 80 characters here.**

The reason this message is inserted is because an alarm has not been triggered for the affected Notify agent. Only on an alarm is the message generated which can then be substituted into this macro. This message is substituted instead to account for the worst-case scenario regarding the length of the alarm message macro – maximum size of 80 characters.

2. I cannot connect to my Gmail account. (Authentication error)

Google has updated its security policy regarding 3rd party applications with access to your Google account. You will need to setup an App Password with 2-Step Verification to allow the Adroit Email Protocol Driver to have access to your Google account.

For more information, please refer to this Adroit Knowledge Base article, which can also be viewed in [Appendix A: Gmail Setup for Email Drivers using IMAP \[New Google Security Policy\]](#):

<https://adroittechnologiesautomation.com/t/sending-e-mail-with-smtp-using-gmail/1287>

Or visit Google Help Centre for more information:

<https://support.google.com/accounts/answer/185833?hl=en>

3. I cannot connect to my email account. (Authentication error)

If you are experiencing issues trying to connect to your email address, please confirm the following is correct:

1. Email address (check formatting, characters used)
2. Email password
3. FROM field in the device configuration

If this error has still not been resolved, there may be an issue with the authentication. Similar to Gmail and Google's security policy changes (refer to the previous troubleshooting scenario for more details), there may be a security policy change surrounding 3rd party application access to your email account. If this is the case or if there is no reference of this security policy change, please report it to Adroit Support.

4. I get any of the following errors:

In the Datascope:

ERROR: Email exceeds RAWBYTES max size in _____ field...

In the email notification:

[ADROIT ERROR] This message indicates that your email is too long. You will experience missing or incomplete text or information, and there will be no attachments. Please refer to the Adroit Email driver documentation for more information on how to resolve this, or please contact Adroit Support.

This indicates that the number of characters in the email generated exceeded the maximum size. Adroit recommends that you reduce the number of characters in your email. This can be done by:

- Reducing the number of recipients.
- Reducing the number of macros.
- Simplify the text in the email text to summarise the alarm.
- Send email attachments only if necessary.

If this is not possible, another solution is to change the **RAWBYTESMaxSize** registry setting.



WARNING

Incorrectly configuring this registry setting can cause undesirable behaviour. Please only change this setting if there is no other solution. This setting will affect all drivers that utilise the Rawbytes slot in Adroit. **This setting is only available in Adroit 10.0.5.2 Update 1 or later.** Do not add this registry setting manually.

Computer\HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Adroit Technologies\Adroit\Scanning

Registry Editor

File Edit View Favorites Help

Computer\HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Adroit Technologies\Adroit\Scanning

Name	Type	Data
(Default)	REG_SZ	(value not set)
AllowFastAsPossible	REG_DWORD	0x00000000 (0)
AllowTimestampChangeUpdates	REG_DWORD	0x00000000 (0)
AllowWritesOnStandbyShadow	REG_DWORD	0x00000000 (0)
DumpScanCfg	REG_DWORD	0x00000000 (0)
HaltOnStartupFailure	REG_DWORD	0x00000000 (0)
IntelligentScanning	REG_DWORD	0x00000000 (0)
MarkBadIfDeviceStopped	REG_DWORD	0x00000001 (1)
MarkBadIfWriteFails	REG_DWORD	0x00000001 (1)
MarkBadOnStartup	REG_DWORD	0x00000001 (1)
MonitorStatus	REG_DWORD	0x00000000 (0)
RAWBYTESMaxSize	REG_DWORD	0x00000200 (512)
ThreadStackSize	REG_DWORD	0x00000000 (0)
UseBulkPutSlots	REG_DWORD	0x00000001 (1)
WriteThreadStackSize	REG_DWORD	0x00040000 (262144)

7. Appendix

7.1. Appendix A: Gmail Setup for Email Drivers using IMAP [New Google Security Policy]

This knowledge base article details the steps required in configuring the Gmail IMAP server with the mentioned Adroit email drivers, in line with Gmail's new security policies implemented as of **30 May 2022**. For further information, refer to the following Google Account Help article: <https://support.google.com/accounts/answer/6010255?hl=en>

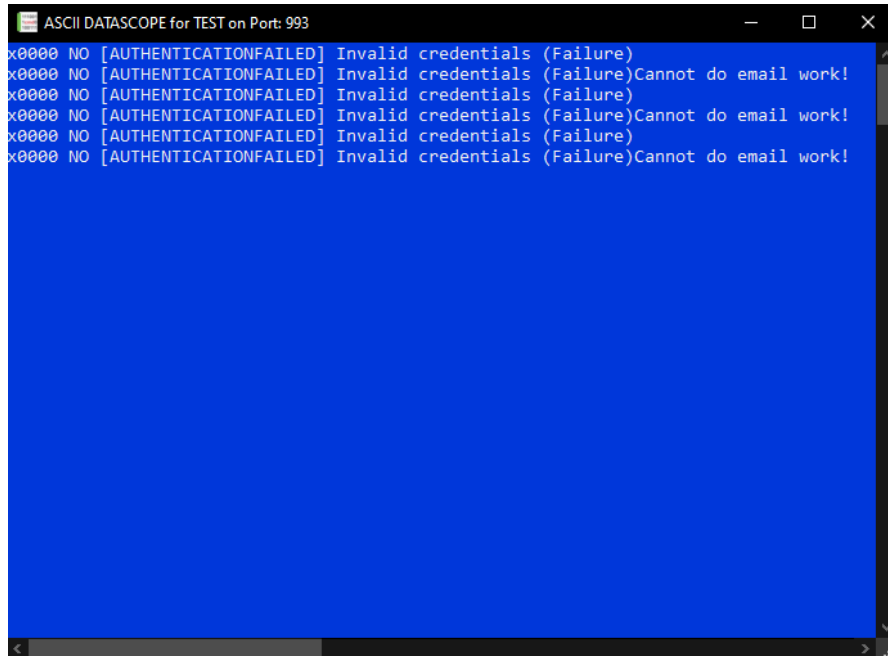
The opening article states the following:

To help keep your account secure, from **May 30, 2022**, Google no longer supports the use of third-party apps or devices which ask you to sign in to your Google Account using only your username and password.

Important: This deadline does not apply to Google Workspace or Google Cloud Identity customers. The enforcement date for these customers will be announced on the Workspace blog at a later date.

For more information, continue to read.

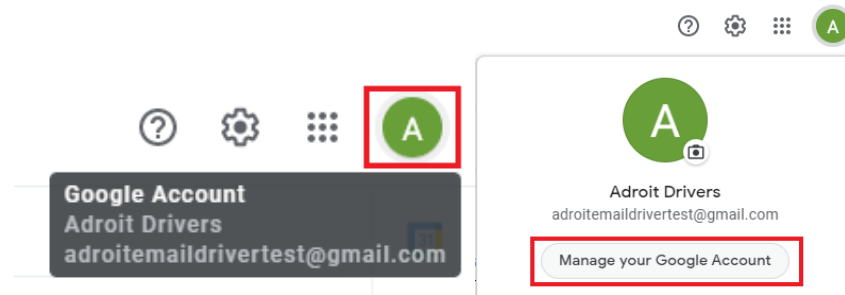
Previously, Adroit email driver's required full access to your Gmail account, and subsequently your Google account, to allow access into your inbox using the Gmail IMAP server. This required adjusting a security setting on your Google account to allow 3rd party applications to access your account. If you have noticed that Adroit has not been able to successfully connect to the Gmail IMAP server, typically stating an authentication error, the problem could be related to this issue.



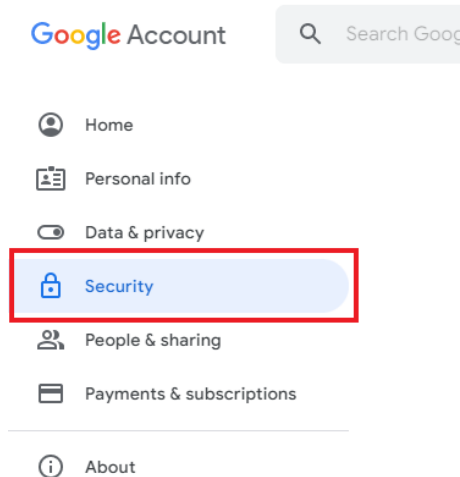
Example of MYCITY Email Client Protocol Driver device authentication error [x0000 NO [AUTHENTICATIONFAILED] Invalid Credentials (Failure) Cannot do email work!]

To access the setting in question, follow the steps provided:

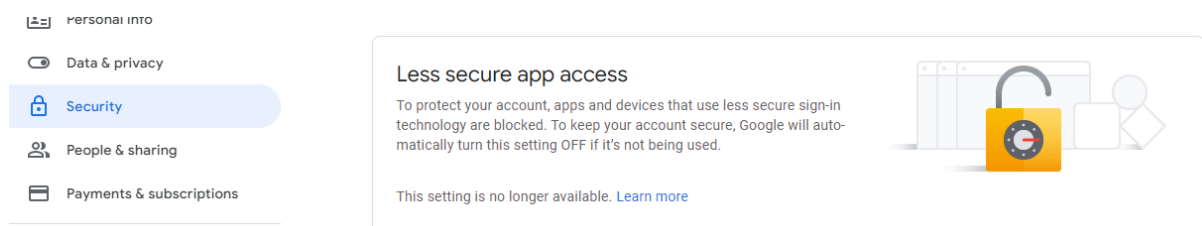
- 1) Accessing from Gmail directly:
 - a. **Log in** to your Gmail account
 - b. Click on the Google Account indicator in the top right-hand corner of the screen. Click on **Manage your Google Account**



c. On the page that opens, click on the **Security** tab located on the left-hand side



d. Under the Security section, there used to be a section **Less secure app access** which was required to be **On**. However, this setting has since been removed

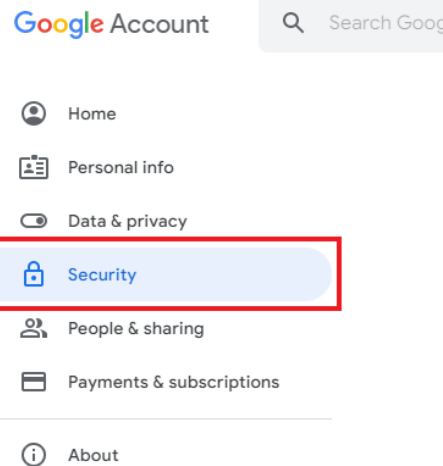


Please Note: if you are unable to see this setting, you probably have **2-Step Verification** turned on. In the past, Adroit would require this be turned off, but please read further as this is now required to be on.

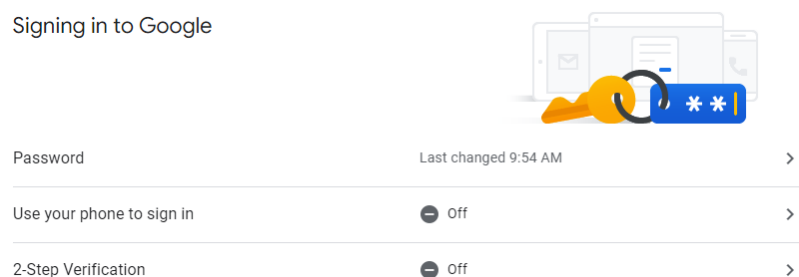
- 2) Accessing account security directly
 - a. **Log in** to your Google Account
 - b. Access your **Google Account**: <https://myaccount.google.com/>
 - c. Follow from Step 1)c outlined previously

To fix this issue, the user will need to create an App Password that Adroit will then use for authentication with Gmail. This requires that 2-Step Verification is enabled. The steps provided below are summarised and can also be read from here, should you require additional information: <https://support.google.com/accounts/answer/185833?hl=en#:~:text=Create%20%26%20use%20App%20Passwords>

- 1) **Log in** to your Google/Gmail Account
- 2) Access your **Google Account**: <https://myaccount.google.com/>
- 3) Click on the **Security** tab on the left-hand side of the screen

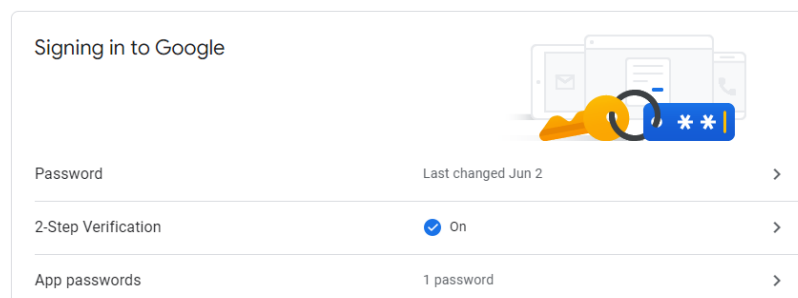


4) Navigate the menu to the section **Signing in to Google**



5) Enable **2-Step Verification** if it is not already. If has not been enabled, do so and follow the prompts by Google. You will need a cellphone number to continue.

6) After setting up 2-Step Verification, click on the new option under the **Signing into Google** called **App passwords**



7) After verifying it is you, you will see a list of App passwords currently registered. Generate a new App password by clicking the **Select app** dropdown box, and selecting the **Mail** option, then the **Select device** dropdown box, selecting the **Windows Computer** option. Click on the **Generate** button.

Select the app and device you want to generate the app password for.

Mail

Select device

iPhone

iPad

BlackBerry

Mac

Windows Phone

Windows Computer

Other (Custom name)

GENERATE

- 8) A message will appear with a 16-character randomly generated password. Copy this and save it somewhere secure. This is the password you will use for Adroit to access the Gmail account.

Generated app password

Your app password for Windows Computer



How to use it

1. Open the "Mail" app.
2. Open the "Settings" menu.
3. Select "Accounts" and then select your Google Account.
4. Replace your password with the 16-character password shown above.

Just like your normal password, this app password grants complete access to your Google Account. You won't need to remember it, so don't write it down or share it with anyone.

[Learn more](#)

Add your Google account

Enter the information below to connect to your Google account.

Email address
seuresally@gmail.com

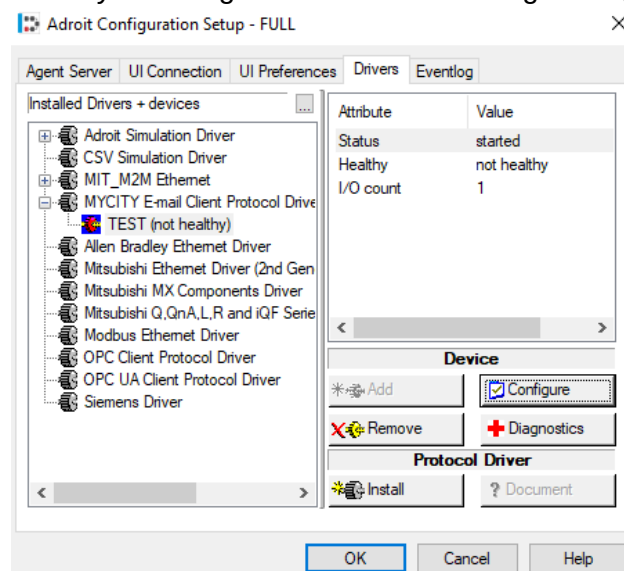
Password

☐ Include your Google contacts and calendars

DONE

- 9) Open the **Classic Adroit Setup** and navigate to the device experiencing the issue.

- 10) Open the device configuration by selecting the device and clicking **Configure**.



- 11) Ensure the details of the device have been input correctly and place the App password previously generated into the **Login Password/Password** input field.

MYCITY E-mail Client Protocol Driver : TEST

Port Selection
☐ Enable Secondary
☒ Primary
☐ Secondary

Local Device Settings
Station Id

OK
Cancel

Common Mail Settings (*Resource key)
Resource

*Mailbox Type
☒ IMAP
☐ POP

*Mail Server
*Mail Port

Login Name
Login Password

Mailbox Poll Period (ms)
☒ Move Used/ToDo Messages to Backup/ToDo Mailboxes.
☐ Delete Used/ToDo Messages.
☐ Delete Spam Messages.
☒ Server uses SSL.
☒ Detailed Debugging on Datascope.

Incoming Mailbox
Backup Mailbox
ToDo Mailbox
Error Mailbox

12) Exit the Device configuration by clicking on **OK**. You may be prompted to change the current resource for all mailboxes if you are using the MyCity Email driver, for example. You can create a new resource or override it for all devices using the same resource.

13) The device should now be marked as healthy. Open the datascope to see if the error is persisting.

Please Note: you may need to **restart** your **Agent Server** before the changes are applied.

Please Note: The App password generated to allow access to Gmail's Email Server does not rely on the verification that 2-Step Verification provides, as it allows the third-party application limited control direct access to your Gmail account. Please keep this password private and secure.

However, considering these changes, Google may continue to further change and evolve their security policies, with or without notice, which can affect the ability of the Adroit email drivers to function as expected. It is therefore recommended that users provide an alternative email with a different provider as a backup solution, or to migrate to another email provider.

Provided is a list of tested email providers that can be utilized with Adroit email drivers as an alternative to Gmail, and more will be added in the future:

- Outlook
-

Please contact Adroit Support should you wish to enquire about compatibility with other email service providers.

For further details on device configuration, please refer to the respective Adroit driver documentation. For any additional information or queries, please feel free to contact [Adroit Support](#) or search our [Knowledge Base](#).

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.2	25-Jan-2023	K. Robinson	Overview section text edits, added Troubleshooting for email authentication errors. Added Appendix A.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
1	2-Feb-2012	First Released Document
4	8-Jan-2013	First official release.
10	9-Dec-2022	Added error checks for email character/size counts related to RawbytesMaxSize registry key.

